

KRISHI ATTRI

✉ krishiattriwork@gmail.com | 💻 [krishi-attri15](https://github.com/krishi-attri15) | 🐙 [Archerkattri](https://Archerkattri.github.io) | 🌐 archerkattri.github.io

EDUCATION

University of Central Florida

Ph.D., Mechanical Engineering — Post Master's Track

ORCGS Doctoral Fellow | REAL Lab (Advisor: Prof. Hwan Choi)

Seoul National University

M.S., Mechanical Engineering | GSFS Scholar | Soft Robotics & Bionics Laboratory

Villanova University

B.S., Mechanical Engineering | Minor in Mechatronics | Control & Dynamics 📄 [Diploma](#)

Orlando, FL, USA

Aug 2026 (Incoming)

Seoul, South Korea

Sept 2024 – June 2026

Villanova, PA, USA

Aug 2020 – May 2024

EXPERIENCE

Graduate Research Student

Sept 2024 – Aug 2026

Soft Robotics & Bionics Laboratory, Seoul National University | Seoul, South Korea

- **Humanoid Robot Project — “Alchemist” Project (MOTIE, Korea):** Contributing to Phase 2 (2024–2026) of a national humanoid-robot R&D program; integrating visuo-tactile SLAM and dexterous in-hand manipulation modules into the full-scale humanoid prototype.
- **GaussianFeels — Object-Centric Gaussian SLAM for Visuo-Tactile In-Hand Manipulation (M.S. Thesis, Advisor: Prof. Yong-Lae Park):** Built an online visuo-tactile reconstruction and 6-DoF pose-tracking system on an explicit object-centric 3D Gaussian Splatting map — contact-aware population management, occlusion-aware multimodal supervision, and a multi-residual Levenberg–Marquardt SE(3) tracker over a frozen Gaussian-density SDF (analytic Jacobian through gsplat), seeded by a Hunyuan3D image-to-3D prior. On FeelSight (14 objects), reached median ADD-S **0.84 mm** sim / **3.4 mm** real at ≈ 28 / ≈ 23.5 FPS — matching or beating model-free NeuralFeels at $\approx 7.6\times$ the frame rate with no supplied CAD model.
- **PoP-SLAM — Neural Point Cloud-Based SLAM:** Co-developed a dense visual SLAM system with a projection-first rendering strategy that eliminates nearest-neighbor searches via GPU-vectorised projection of $\sim 15\,000$ neural points/frame. Achieved **best ATE RMSE of 0.75 cm** on TUM-RGBD (outperforming Point-SLAM, NICE-SLAM, ESLAM) at **~ 4 FPS**. Introduced direct occlusion detection via multi-keyframe depth masking. 📄 [Paper](#)

Robotics & Mechatronics Researcher

Aug 2023 – May 2024

Villanova University | Villanova, PA, USA

- Developed SLAM and visual odometry for quad-wheel robots in GNSS-denied environments via LiDAR–camera fusion, CNN-based feature extraction, and GPU point cloud processing. 📄 [EOD robot](#)
- Built a full ROS-based navigation stack (path planning, obstacle avoidance) with Arduino/Raspberry Pi microcontrollers for real-time control; implemented probabilistic localisation (2D histogram + 1D Kalman filters) and real-time 3D mapping via computer vision.

TECHNICAL SKILLS

SLAM & 3D Reconstruction: 3D Gaussian Splatting, Visuo-Tactile SLAM, Neural Point Cloud SLAM, SfM, RGB-D Reconstruction, Object-Centric Gaussian Mapping

Deep Learning & AI: PyTorch, CUDA (custom kernels), TensorFlow, Multimodal Learning, Differentiable Rendering, CNN / RNN / ViT

Generative 3D, Registration & AI Tooling: HiCache++ (DMD velocity-cache), diffusion/flow caching, Hunyuan3D, TRELIS, SAM 3D, DiT, gsplat, image-to-3D, splat registration (splatreg), MCP & formal-math tooling (mathlas)

Robotics & Middleware: ROS, OpenCV, Open3D, Sensor Fusion, Real-time Inference, NVIDIA Omniverse

Hardware & Sensors: DIGIT Tactile Sensor, RGB/RGB-D Cameras, LiDAR, VectorNav IMU, RTK GPS, UR5e, Allegro Hand, Vicon/OptiTrack

Programming: Python, C/C++, CUDA, MATLAB, Arduino, $\LaTeX$, HTML

Tools & Systems: Linux (Ubuntu), SOLIDWORKS, Blender, Git/GitHub, KiCAD, Jupyter Notebooks

PROJECTS & ACHIEVEMENTS

HiCache++ — Training-Free DMD Velocity-Cache (Diffusion/Flow) 📄 [PyPI](#) | Python, PyTorch, CUDA

2026

- **HiCache++** — a model-agnostic, training-free DMD/exponential velocity-cache (extends the prior HiCache/Hermite + TaylorSeer lines) that cuts redundant denoising compute while staying near-lossless. Built and benchmarked across **6+ backbones** (Hunyuan3D-2/2.1, TRELIS v1/v2, Meta SAM 3D, Fast-SAM3D, DiT-XL/2) as open-source repos. Also released **splatreg** 📄 [PyPI](#) (SE(3)/Sim(3) registration + merge for 3DGS).

Computer Vision Object Detection App | Python, PyTorch, React, Flask

Summer 2024

- Full-stack React/Flask app for real-time image/video object detection; implemented and benchmarked five models (Faster / Mask / Keypoint R-CNN, RetinaNet, SSDlite) in PyTorch and TensorFlow on COCO.

Capstone: Plant Lifting Device for 3D Imaging | FMC Corporation (Sponsored) 📄 [Demo](#) 📄 [Award](#)

Aug 2023 – May 2024

- Led a multidisciplinary team from concept to working prototype (actuators, motion systems, wiring, waterproofing); **1st Place, Most Innovative Solution** at the capstone showcase.

CERTIFICATIONS & AWARDS

NVIDIA Computer Vision Nanodegree (Udacity, 2024) · **K-MOOC Soft Robotics & Applications** (SNU, 2024) · **Robotics Specialisation** (Coursera/UPenn, 2021)
GRE: Quant 166 (80th pctl), Verbal 153 · Dean's List: Fall 2020, Spring 2021 · Capstone 1st Place (2024) · ICE Competition 3rd Place (2020)